Hacking Wireless Networks

CEH v13 — Module 16 Practical Lab Report

Made by Dhruva

Candidate	Dhruva Khare	**Course**	CEH v13 (Certified Ethical Hacker)
Module	16 – Hacking Wireless Networks	**Date**	October 21, 2026
Lab Duration	14 minutes, 44 seconds	**Result**	**2 / 2 — PASSED**

1. Objective

This report documents the completion of CEH v13 Module 16 (Hacking Wireless Networks), covering two practical exercises: (1) analysis of captured 802.11 wireless traffic using Wireshark, and (2) recovery of a WPA2 pre-shared key from a captured handshake using the Aircrack-ng suite. All activity was performed in an isolated lab environment using instructor-provided sample capture files, strictly for certification training purposes.

2. Tools & Environment

Tool	Purpose	Version / Notes
Wireshark	Wi-Fi packet capture analysis	Used to inspect 802.11 frames
Aircrack-ng	WPA2 handshake key recovery	Aircrack-ng 1.7
Sample files	WPA2crack-01.cap + wordlist	Provided under CEHv13 Module 16 Sample Captures

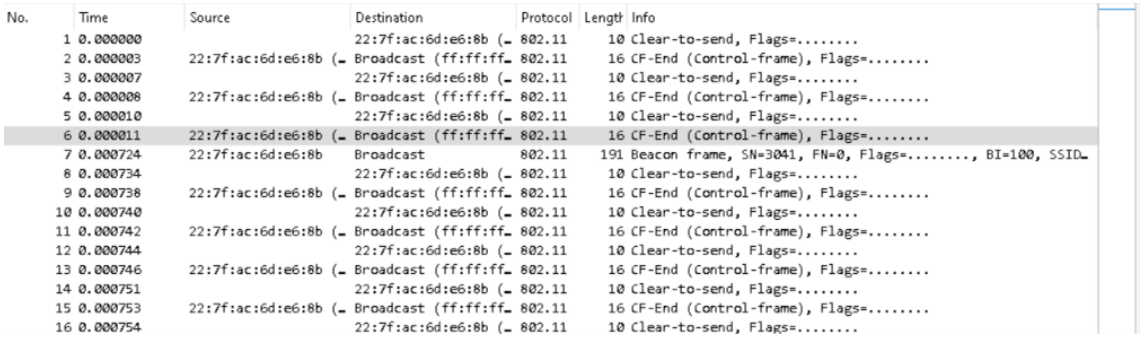
3. Exercise 1 — Wireshark Wi-Fi Packet Analysis

Task

Use the Wi-Fi packet-sniffing tool Wireshark to analyze captured Wi-Fi packets (WPA2crack-01.cap) and identify the protocol that indicates the wireless packets.

Procedure

- Opened the sample capture file WPA2crack-01.cap in Wireshark.
- Reviewed the frame listing, including Control-frame exchanges (Clear-to-Send, CF-End) and a Beacon frame broadcast (SN=3041, BI=100).
- Inspected the Protocol column, which consistently reported traffic as 802.11, confirming the captured packets are IEEE 802.11 wireless frames (management and control frames such as Beacon, CTS and CF-End).



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000		22:7f:ac:6d:e6:8b (…	802.11	10	Clear-to-send, Flags=........
2	0.000003	22:7f:ac:6d:e6:8b (…	Broadcast (ff:ff:ff…	802.11	16	CF-End (Control-frame), Flags=........
3	0.000007		22:7f:ac:6d:e6:8b (…	802.11	10	Clear-to-send, Flags=........
4	0.000008	22:7f:ac:6d:e6:8b (…	Broadcast (ff:ff:ff…	802.11	16	CF-End (Control-frame), Flags=........
5	0.000010		22:7f:ac:6d:e6:8b (…	802.11	10	Clear-to-send, Flags=........
6	0.000011	22:7f:ac:6d:e6:8b (…	Broadcast (ff:ff:ff…	802.11	16	CF-End (Control-frame), Flags=........
7	0.000724	22:7f:ac:6d:e6:8b	Broadcast	802.11	191	Beacon frame, SN=3041, FN=0, Flags=........, BI=100, SSID…
8	0.000734		22:7f:ac:6d:e6:8b (…	802.11	10	Clear-to-send, Flags=........
9	0.000738	22:7f:ac:6d:e6:8b (…	Broadcast (ff:ff:ff…	802.11	16	CF-End (Control-frame), Flags=........
10	0.000740		22:7f:ac:6d:e6:8b (…	802.11	10	Clear-to-send, Flags=........
11	0.000742	22:7f:ac:6d:e6:8b (…	Broadcast (ff:ff:ff…	802.11	16	CF-End (Control-frame), Flags=........
12	0.000744		22:7f:ac:6d:e6:8b (…	802.11	10	Clear-to-send, Flags=........
13	0.000746	22:7f:ac:6d:e6:8b (…	Broadcast (ff:ff:ff…	802.11	16	CF-End (Control-frame), Flags=........
14	0.000751		22:7f:ac:6d:e6:8b (…	802.11	10	Clear-to-send, Flags=........
15	0.000753	22:7f:ac:6d:e6:8b (…	Broadcast (ff:ff:ff…	802.11	16	CF-End (Control-frame), Flags=........
16	0.000754		22:7f:ac:6d:e6:8b (…	802.11	10	Clear-to-send, Flags=........

Figure 1. Wireshark capture of 802.11 frames in WPA2crack-01.cap.

Protocol Identified	802.11

Score: 1 / 1 — Correct

4. Exercise 2 — WPA2 Key Recovery with Aircrack-ng

Task

Use the Aircrack-ng suite to crack a WPA2 network and recover the pre-shared key, using the sample captured Wi-Fi packets and wordlist provided at `/home/attacker/Desktop/CEHv13 Module 16 Hacking Wireless Networks`.

Procedure

- Launched Aircrack-ng 1.7 against the captured handshake file, pointing it at the supplied dictionary wordlist.
- Aircrack-ng performed a dictionary attack, testing candidate keys against the captured WPA2 4-way handshake at a rate of approximately 205 keys/second.
- After testing 9 of 481 candidate keys, the tool reported KEY FOUND, deriving the Pairwise Master Key (PMK), Pairwise Transient Key (PTK) and the EAPOL HMAC used to validate the match.

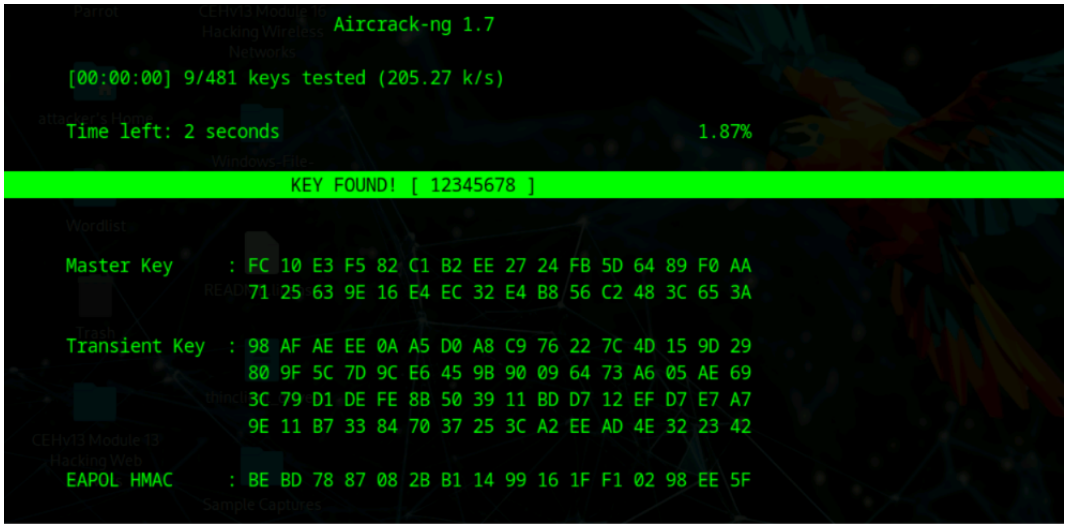


Figure 2. Aircrack-ng dictionary attack result against the WPA2 handshake.

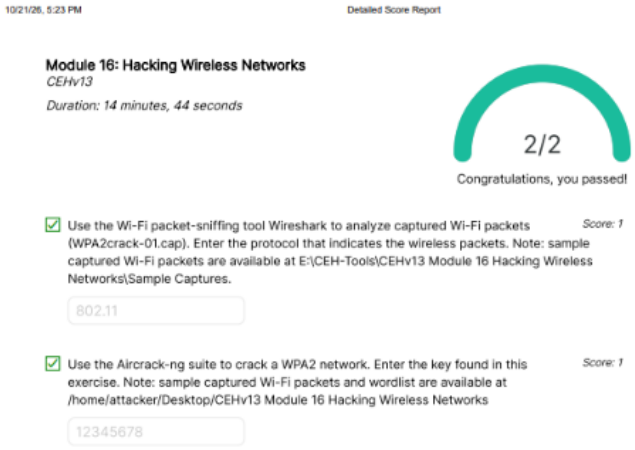
WPA2 Pre-Shared Key Recovered	12345678

Score: 1 / 1 — Correct

Note: the recovered key is a weak, numeric-sequence password. This illustrates why WPA2 networks remain vulnerable to dictionary/brute-force attacks when a weak pre-shared key is used, reinforcing the need for long, high-entropy passphrases (or WPA3-SAE) in production wireless deployments.

5. Lab Score Summary

The official lab scoring report confirms successful completion of both exercises in Module 16, with a final score of 2 out of 2 (100%).


10/21/26, 5:23 PM Detailed Score Report

Module 16: Hacking Wireless Networks
CEHv13
Duration: 14 minutes, 44 seconds

2/2
Congratulations, you passed!

Use the Wi-Fi packet-sniffing tool Wireshark to analyze captured Wi-Fi packets (WPA2crack-01.cap). Enter the protocol that indicates the wireless packets. Note: sample captured Wi-Fi packets are available at E:\CEH-Tools\CEHv13 Module 16 Hacking Wireless Networks\Sample Captures. Score: 1

802.11

Use the Aircrack-ng suite to crack a WPA2 network. Enter the key found in this exercise. Note: sample captured Wi-Fi packets and wordlist are available at /home/attacker/Desktop/CEHv13 Module 16 Hacking Wireless Networks Score: 1

12345678

Figure 3. Official CEHv13 Module 16 detailed score report.

#	Exercise	Answer Submitted	Score
1	Identify protocol in Wireshark capture	802.11	1 / 1
2	Crack WPA2 key with Aircrack-ng	12345678	1 / 1
	Total		**2 / 2 (100%)**

6. Key Takeaways

• 802.11 is the IEEE standard governing Wi-Fi frames at the MAC/PHY layer; Wireshark can decode management, control and data frame subtypes directly from a raw capture.

• A captured WPA2 4-way handshake, combined with a weak or predictable pre-shared key, can be broken quickly offline using a dictionary attack in Aircrack-ng.

• Weak numeric or dictionary-word Wi-Fi passphrases (e.g. “12345678”) provide little real-world protection and should be replaced with long, random passphrases or WPA3-SAE where supported.

• This exercise was performed entirely against instructor-supplied sample capture files in an isolated training lab, consistent with authorized, ethical penetration-testing practice.

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